SHAKE-derived scalar batch throughput for affine rotation tables

This report compares the ordinary-affine-layer and full two-x table paths on the same 12,288 distinct SHAKE256-derived secp256k1 scalars. It is a Tier 1 cloud-instance batch-throughput measurement; it does not measure one-call latency or establish physical-host isolation.

Inputs, implementation, and checks

The fixture seed is SHA-256 of the frozen 128-point base-fixture digest under the domain `prime-j0-uniform-scalar-v1`. A SHAKE256 stream over an increasing 64-bit counter is rejected unless the 32-byte value lies in `1…n−1` and has not appeared before. Each accepted point is computed with independent Python binary scalar multiplication and checked on the secp256k1 curve. The generator used 12,288 counters for 12,288 unique scalars; the committed fixture hash is `a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3`.

The first fixture generation completed its computation but could not write to a full local SSD. That failed attempt and the successful rerun are retained. The hash-guarded native overlay adds two benchmark-only input modes to the same binary, so the two table formats use identical source, executable, and fixture. All outputs are verified against independently generated points after timing.

The independent auditor regenerated the fixture and checked provenance archives, binary/source/fixture hashes, all 96 raw arms and 48 pairs, each recorded point, each adjacent A/A gate, and the pre-declared bootstrap result. Running `analyze_tier1_uniform.py` reproduced `tier1-affine-uniform-cpu40-audit.json` byte-for-byte.

Measured result

Same-input batch measure	Ordinary affine	Full two-x
Mean online time, 12,288 outputs	70.737837 ms	62.137246 ms
Mean table preparation	3,950.829 ms	3,967.026 ms
Retained native table	26,035,988 bytes	37,192,948 bytes

The median ordinary-affine/full-two-x online-time ratio was 1.155612×, with repeat-bootstrap 95% interval [1.101903, 1.178301] across 16 comparison repetitions. Every A/A control

passed its frozen [0.90, 1.11] gate. The table preparation interval is reported separately from online evaluation, and the full two-x table retains 11,156,960 additional bytes. The timing covers 96 batches of 128 scalar multiplications and affine recovery after fixture loading and table construction; correctness checks follow the timed interval.

The result is scoped to this observed RunPod Tier 1 instance and this batch input law. The container records process placement and cgroup/host observations but does not prove physical-host exclusivity. Single-scalar cold latency and other scalar distributions remain unmeasured here.

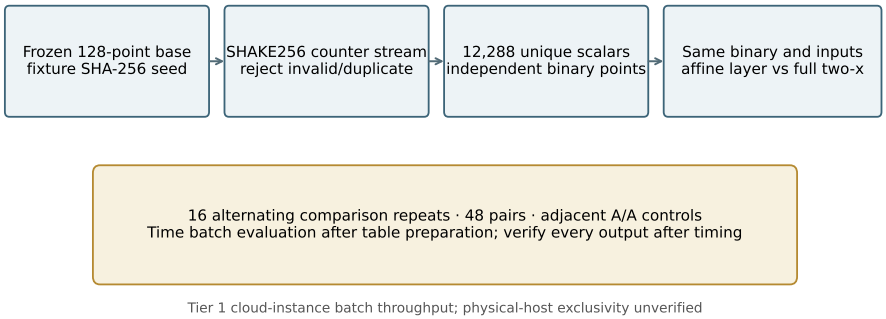


Figure 1: Fixture construction and panel flow

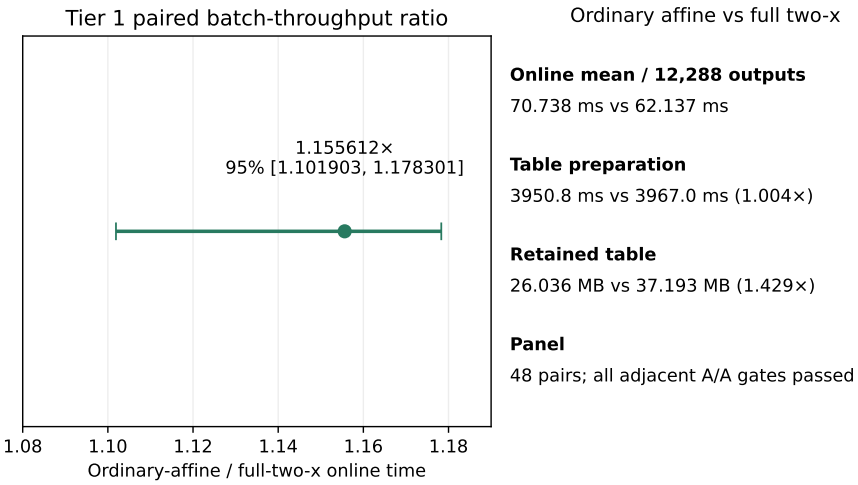


Figure 2: Measured Tier 1 throughput ratio

Visual source and graph scope

`visuals.py` generates the editable source graphs `fixture-and-panel.svg` and `tier1-throughput.svg`, plus PDF renderings. The graph set records the measured scalar-batch comparison and fixture flow. The existing curve/isogeny catalog graphs were checked; this experiment uses the existing secp256k1 curve and adds no curve, subgroup, map, or route fact.

The source-linked report is this Markdown file and the committed `REPORT-2026-10-19.pdf`. Rebuild with:

```
MPLCONFIGDIR=/tmp/mpl-cache XDG_CACHE_HOME=/tmp/font-cache python3 visuals.py
SOURCE_DATE_EPOCH=0 pandoc REPORT-2026-10-19.md --resource-path=.:figures \
  --pdf-engine=xelatex -V mainfont="DejaVu Serif" \
  -V monofont="DejaVu Sans Mono" -o REPORT-2026-10-19.pdf
```

Raw arms, pair ledgers, fixture, native provenance, and the independent audit remain linked from `README.md`.